CENTRAL UNIVERSITY OF KARNATAKA

HELIOPATH

A Smart Solar Footpath for Sustainable Campus Mobility.

Project Report – Prototype 3

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1. Introduction

The increasing demand for clean and sustainable energy systems has encouraged the development of smart infrastructure solutions capable of supporting renewable energy integration and eco-friendly transportation systems. HELIOPATH is a smart solar footpath infrastructure project designed to integrate solar energy generation, pedestrian pathways, EV charging systems, and sustainable campus mobility into a single modular infrastructure system.

The project aims to utilize unused footpath and pedestrian pathway spaces for solar energy generation while simultaneously providing shaded walkways, charging support, and green transportation facilities inside the campus environment.

Prototype 3 represents the refined and improved version of the HELIOPATH project after analyzing the limitations and challenges observed in previous prototypes.

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2. Objective of Prototype 3

The main objective of Prototype 3 is to create a practical, modular, and campus-oriented smart solar footpath system capable of supporting renewable energy generation and sustainable transportation infrastructure.

The prototype focuses on:

- Solar energy generation using modular solar canopy systems
- Footpath-based infrastructure deployment
- EV charging integration
- Battery
- Battery energy storage integration
- Cycle parking and green mobility support
- Smart campus connectivity between hostel and academic areas

- Realistic and modular infrastructure planning

Another important objective of Prototype 3 is to improve the structural organization, realism, and practical implementation capability compared to the previous prototypes.

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3. Prototype Description

The layout planning of Prototype 3 was developed after analyzing the limitations of previous models. A vertical campus-based layout structure was selected to improve spacing, road flow, and realistic infrastructure organization.

The Academic Building was positioned at the center of the layout to act as the primary infrastructure and energy distribution zone. The Boys Hostel and Girls Hostel were connected through curved road and footpath pathways to improve campus mobility and accessibility.

Special attention was given to:

- Curved road alignment
- Footpath placement
- Solar canopy positioning
- EV charging accessibility
- Battery storage location
- Cycle stand distribution

The curved road structure improved the realism of the model and allowed better integration of modular solar structures along the pedestrian pathways.

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4. 2D Layout Planning of Prototype 3

To improve the overall organization, realism, and practical implementation of the HELIOPATH project, a detailed 2D layout was prepared before refining the final CAD model of Prototype 3. The purpose of creating the 2D layout was to visualize the campus road network, pedestrian pathways, building placement, and solar footpath integration in a more systematic and structured manner.

The 2D layout of Prototype 3 includes the Boys Hostel, Girls Hostel, Academic Building, CS Department, and Physiology Department connected through a curved road and footpath network. The pathways were designed to simulate realistic campus movement patterns and to identify suitable locations for the implementation of the HELIOPATH modular solar footpath system.

Compared to the previous prototypes, Prototype 3 focuses more on practical campus deployment, modular infrastructure planning, and realistic sustainable mobility integration. The implementation of curved road and footpath structures improved the visual appearance and allowed better integration of solar canopy systems along pedestrian pathways.

The prepared layout also helped in understanding the spacing and positioning requirements for:

- Solar canopy structures
- EV charging stations
- Battery Energy Storage System (BESS)
- Cycle parking infrastructure
- Smart lighting systems

The following dimensions were considered during the preparation of the Prototype 3 layout:

- Base size: 1200 mm × 1700 mm
- Road width: 90 mm
- Footpath width: 30 mm
- Approximate building block size: 100 mm × 100 mm

The actual campus distances considered during the planning stage are:

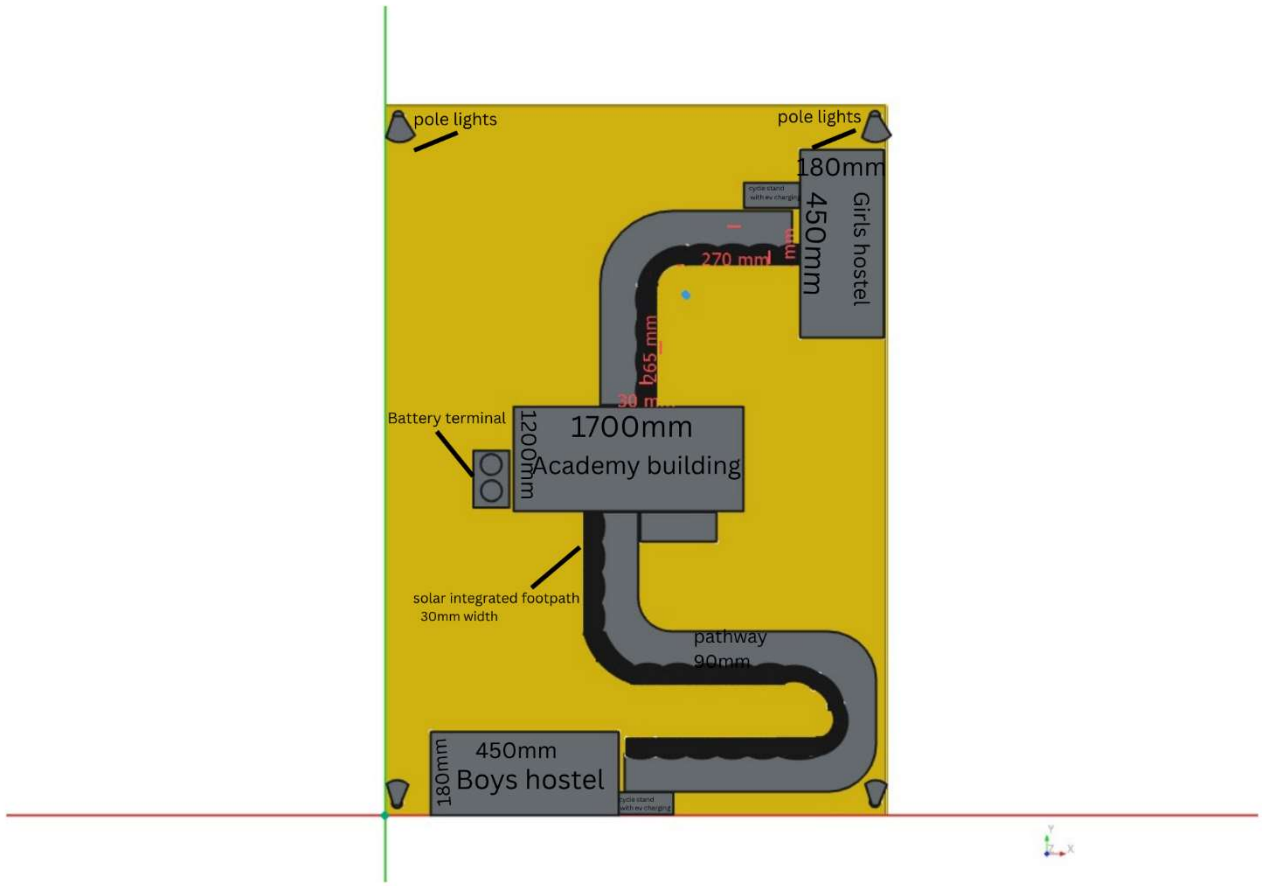
- Boys Hostel to Physiology Department ≈ 750 m
- CS Department to Girls Hostel ≈ 650 m

The 2D planning stage played an important role in reducing design complexity and identifying structural and spacing-related issues before continuing toward detailed CAD development. It also helped improve the alignment of roads, placement of buildings, solar canopy positioning, and the integration of sustainable infrastructure components.

This layout serves as the foundational stage for further refinement and future development of the HELIOPATH smart solar footpath system, including:

- Improved building structures
- Better solar canopy alignment
- Enhanced campus mobility planning
- Efficient renewable energy integration
- Smart sustainable infrastructure implementation

Figure (i): 2D Campus Layout of Prototype 3



5. Solar Canopy System

The solar canopy system is the main feature of Prototype 3. The canopy structures are placed along the pedestrian footpath and are supported through modular repeated structural segments.

Each module contains:

- Vertical support pillars
- Curved roof support beam
- Solar panel mounting section

The modular approach allows easy scalability, easier maintenance, and improved structural organization. Instead of using one continuous roof structure, multiple smaller modules were used to improve practical implementation and curved path adaptability.

The solar canopy system supports renewable energy generation while also providing shaded pathways for pedestrians during hot weather conditions.

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6. Battery Energy Storage System (BESS)

A centralized Battery Energy Storage System (BESS) was integrated near the Academic Building to store the generated solar energy.

The battery system can support:

- EV charging systems
- Smart lighting systems
- Campus utility loads
- Future renewable integration systems

The centralized placement improves accessibility, maintenance capability, and energy distribution efficiency throughout the campus infrastructure.

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7. EV Charging and Cycle Infrastructure

Prototype 3 includes dedicated EV charging stations placed near the Academic Building and major pedestrian zones to improve accessibility for students and campus users.

Cycle parking stands were also integrated near hostel zones and academic areas to promote green transportation and reduce dependence on fossil-fuel-based transportation systems.

The inclusion of cycle infrastructure represents an important improvement over Prototype 1.

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8. Challenges Faced During Development

Several challenges were encountered during the development of Prototype 3, including:

- Curved pathway implementation
- Solar structure placement along curved paths
- Modular repetition alignment
- CAD body and clone management issues
- Structure scaling and spacing adjustments
- File management and geometry reconstruction challenges

Multiple redesigns and refinements were performed to improve realism, structural balance, and implementation practicality.

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9. Future Improvements

Several improvements are planned for future development, including:

- Improved building detailing and architectural realism
- Better solar panel rendering and textures
- Smart IoT-based monitoring systems
- Automated lighting integration
- Advanced drainage system integration
- Realistic landscape elements such as trees and benches
- Improved road markings and pathway detailing

Future versions may also include real-time energy monitoring and smart campus management systems.

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10. Conclusion

Prototype 3 represents a major improvement in the development of the HELIOPATH project. The prototype successfully demonstrates the integration of renewable energy infrastructure, pedestrian mobility systems, EV charging facilities, and smart campus planning into a single modular infrastructure model.

The project helped improve understanding of CAD modeling, infrastructure planning, modular system design, and sustainable engineering concepts. Despite several technical and modeling challenges, Prototype 3 established a practical and realistic foundation for future development and refinement of the HELIOPATH smart solar footpath system.

11. Methodology

The development methodology of the HELIOPATH Prototype 3 project was divided into multiple stages to ensure systematic planning, layout refinement, and practical infrastructure integration.

11.1 Concept Development

The initial stage focused on identifying the major problems associated with increasing energy demand, fossil fuel dependency, and the lack of sustainable campus mobility infrastructure. The concept of integrating solar energy generation with pedestrian pathways was selected as the primary project direction.

The idea evolved into the development of a smart solar footpath capable of supporting:

- Renewable energy generation
- Pedestrian movement
- EV charging systems
- Battery storage systems
- Green transportation infrastructure

11.2 Analysis of Previous Prototypes

The limitations of Prototype 1 and Prototype 2 were analyzed before starting Prototype 3.

Prototype 1 Limitations

- Large road coverage structure
- Complex implementation
- Lack of cycle infrastructure
- Less practical deployment approach

Prototype 2 Limitations

- Congested layout planning
- Weak building structure design
- Improper road proportion
- Unorganized infrastructure spacing

These observations helped improve the planning and organization of Prototype 3.

11.3 Layout Planning

A rough hand-drawn layout was first prepared to understand:

- Building positioning
- Road flow
- Footpath connectivity
- Solar structure placement

The campus layout was planned vertically to improve spacing and movement flow. The Academic Building was placed at the center, while the Boys Hostel and Girls Hostel were positioned on opposite sides and connected through curved pathways.

11.4 CAD Modeling

The CAD model was developed using FreeCAD software.

The modeling process included:

- Base platform creation
- Curved road design
- Footpath implementation
- Building placement
- Solar canopy modeling
- Battery storage integration
- EV charging placement
- Cycle stand implementation

Different FreeCAD workbenches such as Sketcher, Draft, and Part Design were used during the modeling process.

11.5 Solar Canopy Development

The solar canopy system was developed using a modular design approach. Instead of implementing one continuous structure, repeated smaller solar modules were designed and placed along the curved pathways.

Each module consisted of:

- Support pillars
- Curved beam structures
- Solar panel mounting surfaces

The modular approach improved scalability and simplified implementation along curved pathways.

11.6 Infrastructure Integration

Additional smart infrastructure systems were integrated into the prototype, including:

- Battery Energy Storage System (BESS)
- EV charging stations
- Smart lighting poles
- Cycle parking stands

The Battery Storage System was placed near the Academic Building to provide centralized energy distribution and accessibility.

11.7 Documentation and Refinement

Throughout the project development process:

- CAD screenshots were recorded
- Prototype images were documented
- Design changes were analyzed
- Multiple refinements were performed

The final model was optimized to improve:

- realism
- modularity
- spacing
- practical implementation capability

12. References

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